AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 5 · WORKBOOK

Authentication and authorisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Authentication and authorisation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Configuration and secrets

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Architecture and data flow

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Authentication identifies an actor; **authorisation** decides whether that actor may perform a particular operation on a particular resource.

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Authentication and authorisation trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Authentication** and **authorisation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Authentication and authorisation explanation against the supplied facts before discussing it.

Use the worked case for **Authentication** and **authorisation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Authentication and authorisation facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Authentication and authorisation facts and keep your reasoning.

A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Authentication and authorisation facts and keep your reasoning.

A report omits this qualification: Successful **authentication** does not grant every resource or operation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Authentication and authorisation facts and keep your reasoning.

Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Authentication and authorisation facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Authentication and authorisation to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **authentication** and **authorisation evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **authentication** and **authorisation**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Authentication and authorisation

> A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C. Explain the labelled step before checking the changed condition.

A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.

Trusted policy grants volunteers read access to C; only curators may edit it.

AI ACADEMY / AI-6/5

Authentication and authorisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify Jo

Authentication establishes Jo's identity and volunteer role.



2. Check the action

Editing C requires curator authority.



3. Deny the edit

Keep C unchanged; the authenticated volunteer lacks this permission.

CHANGE ONE THING

Use a verified curator permitted to edit C.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why Jo’s valid volunteer login does not permit editing C. Which **trusted role** change allows the action, and why is a role written in the request insufficient?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep C unchanged; the authenticated volunteer lacks this permission.

1. Check the action

2. Deny the edit

3. Identify Jo

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Deny the edit Keep C unchanged; the authenticated volunteer lacks this permission. **Authentication** establishes identity.

Authorisation compares that identity with the requested action and resource.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use a verified curator permitted to edit C. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The **authorisation** gate allows the edit, subject to other declared checks.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **authorisation** gate allows the edit, subject to other declared checks. The **trusted role** now matches this operation and object.

Repair a missing building block

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

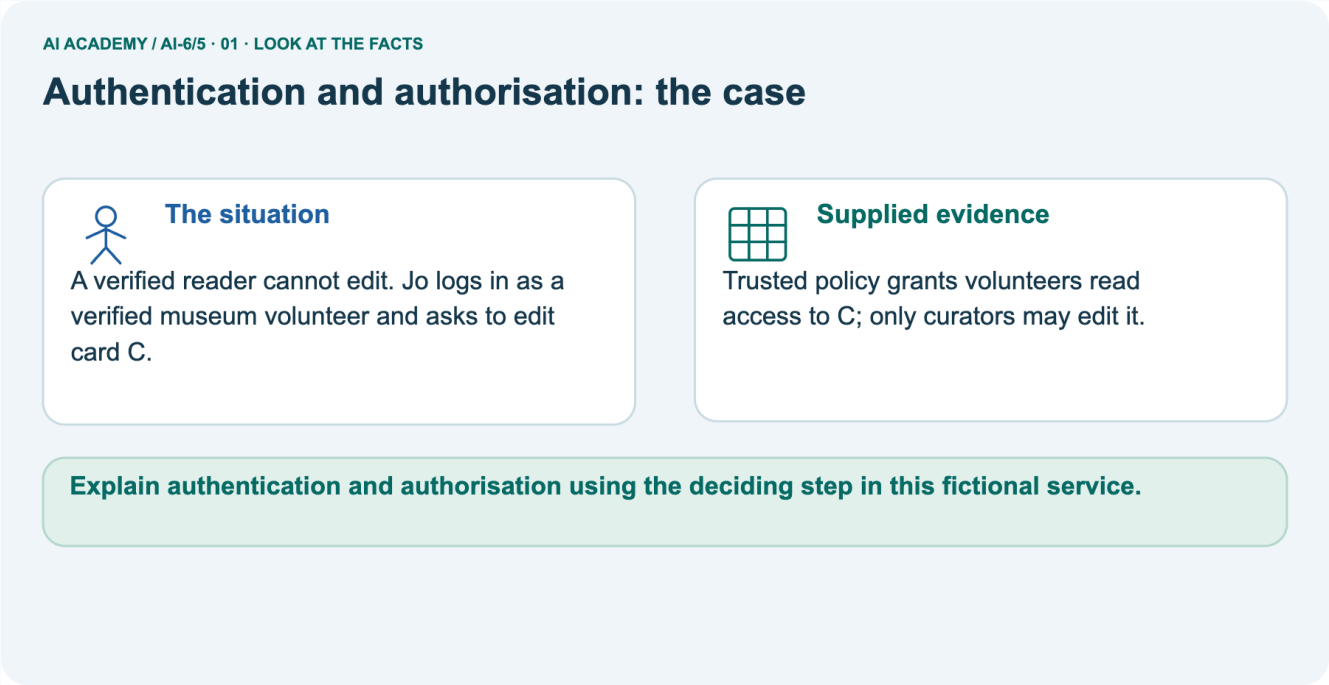
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Authentication and authorisation picture from its labels, then record a relevant limit.

01. Authentication and authorisation: the case



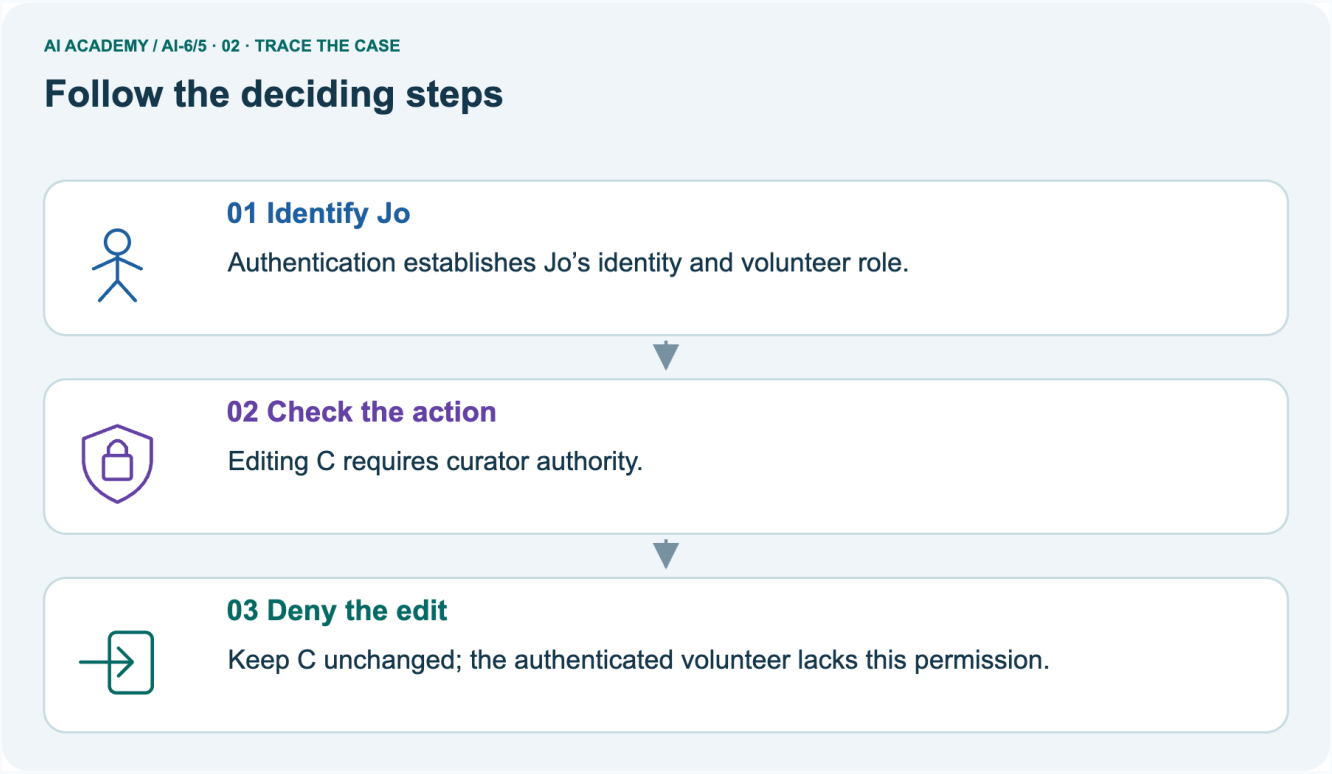
Follow the labelled picture with the matching explanation. Open the vector diagram

A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.
Supplied facts: Trusted policy grants volunteers read access to C; only curators may edit it.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



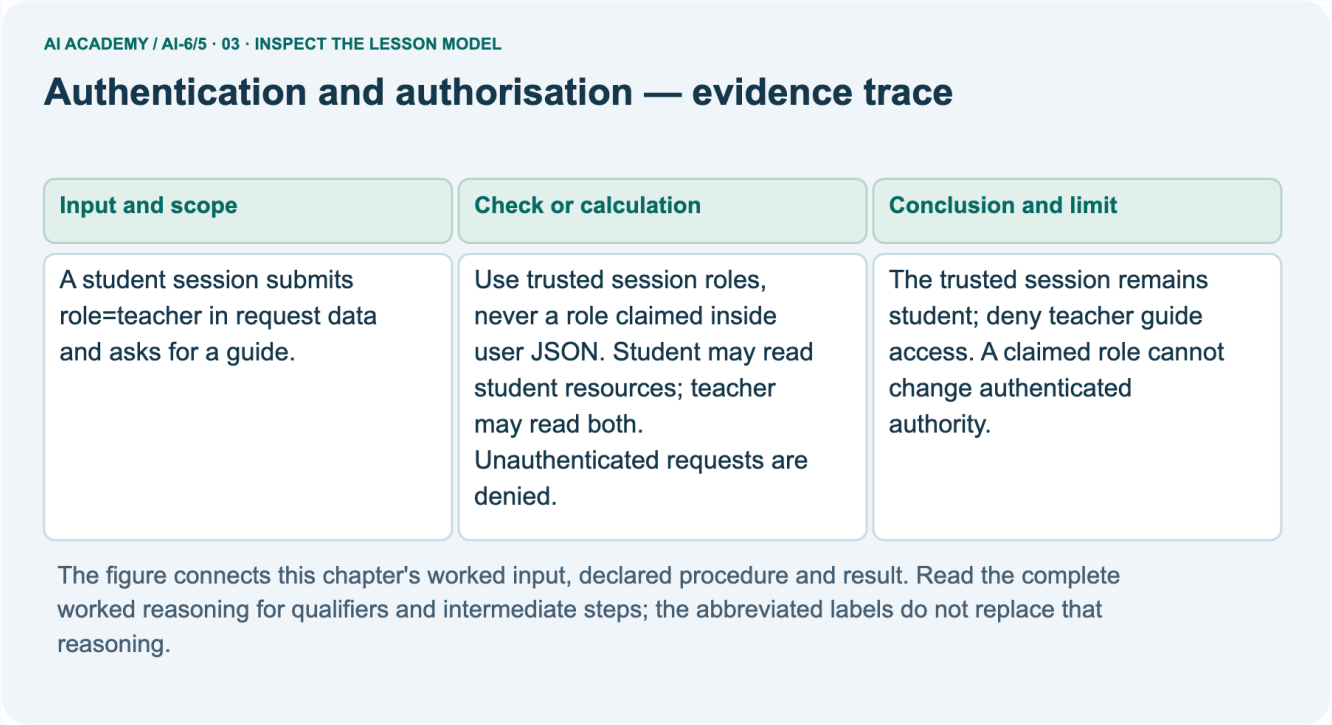
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify Jo: **Authentication** establishes Jo's identity and volunteer role. → Check the action: Editing C requires curator authority. → Deny the edit: Keep C unchanged; the authenticated volunteer lacks this permission.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Authentication and authorisation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

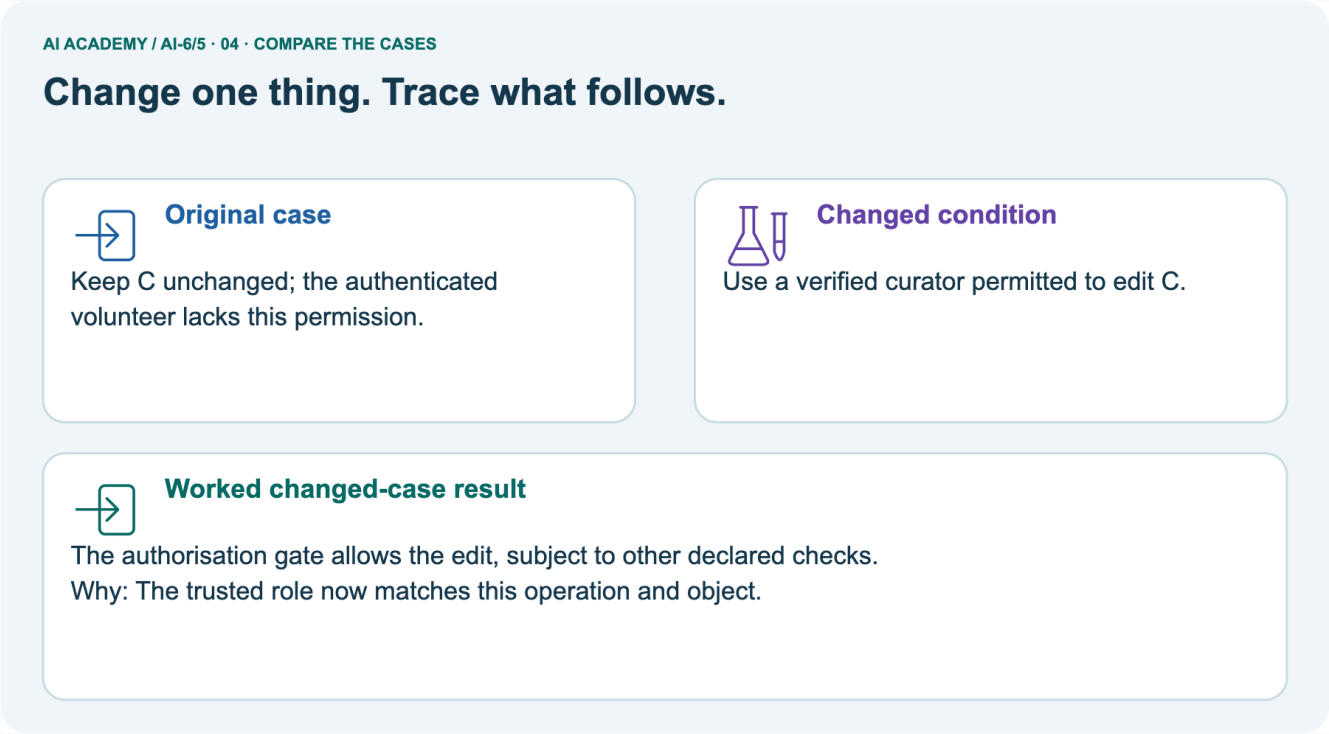
Authentication and **authorisation** — **evidence** trace. A student session submits role=teacher in request data and asks for a guide. | Use trusted session roles, never a role claimed inside user JSON.

Student may read student resources; teacher may read both. Unauthenticated requests are denied. | The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

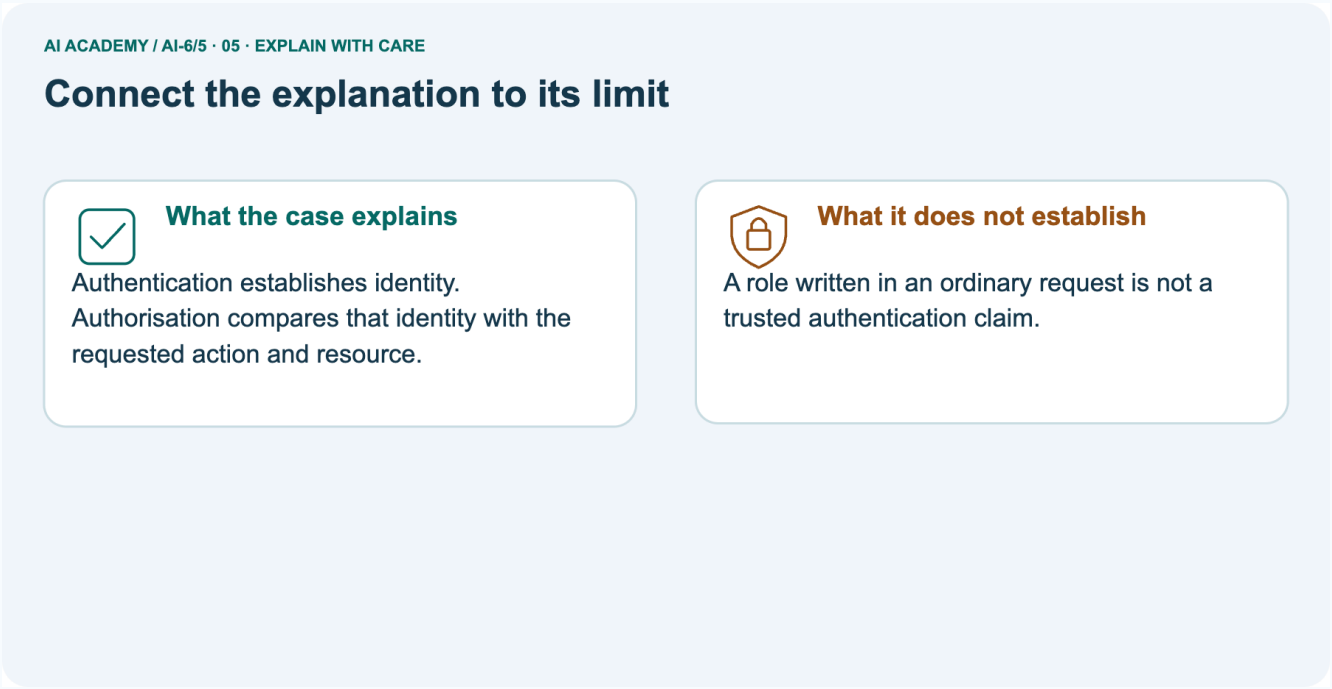
Original case: Keep C unchanged; the authenticated volunteer lacks this permission. Changed condition: Use a verified curator permitted to edit C.

Expected result: The **authorisation** gate allows the edit, subject to other declared checks. Reason: The **trusted role** now matches this operation and object.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



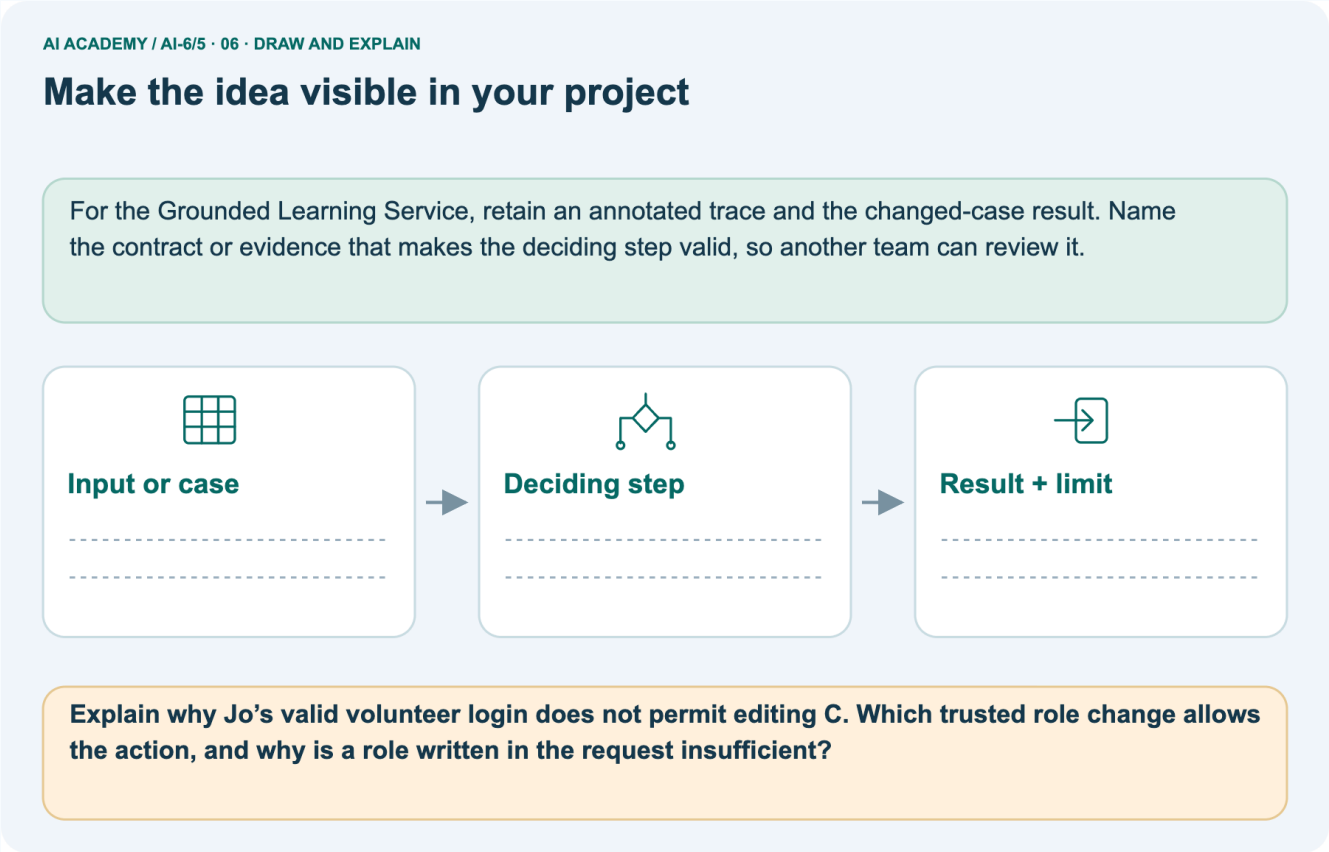
Follow the labelled picture with the matching explanation. Open the vector diagram

Authentication establishes identity. **Authorisation** compares that identity with the requested action and resource. Limit: A role written in an ordinary request is not a trusted **authentication** claim.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why Jo's valid volunteer login does not permit editing C.

Which **trusted role** change allows the action, and why is a role written in the request insufficient?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.